INFORMATION TECHNOLOGY: NOVEMBER 2026 PRACTICAL PAPER 1

SECTION A: SQL ANSWER SHEET

*Type answers **in** the blue boxes*

NAME & SURNAME:

Tinokudaishe Mberi

1.1:

```
SELECT
   tblAIJobs.*
FROM
   tblAIJobs
ORDER BY
   TotalCost DESC;
```

[4]

1.2:

```
SELECT
   ID,
   CompanyName,
   Province,
   GPUHours
FROM
   tblAIJobs
WHERE
   Province = "Gauteng";
```

[3]

1.3:

```
SELECT DISTINCT
   ServiceType
FROM
   tblAIJobs
```

[2]

1.4:

```
SELECT
   TOP 5 ID,
   TotalCost,
   CompanyName
FROM
   tblAIJobs
ORDER BY
   TotalCost DESC;
```

[2]

1.5:

```
SELECT
   ID,
   CompanyName,
   GPUHours,
   tblAIJobs.Status
FROM
   tblAIJobs
WHERE
   GPUHours BETWEEN 120 AND 250
```

[3]

1.6:

```
SELECT
   ID,
   CompanyName,
   tblAIJobs.Status,
   Priority,
   Technician
FROM
   tblAIJobs
WHERE
   (
      (Status = "Queued")
      OR (Status = "Running")
   )
   AND IsNull(Technician)
   AND NOT (Priority = "Low");
```

[5]

1.7:

```
UPDATE tblAIJobs
SET Priority = "High"
WHERE
(RenewablePowered = FALSE)
AND (Province = "Gauteng")
OR (Province = "Western Cape")
```

[7]

1.8:

```
DELETE FROM tblAIJobs
WHERE BookingDate < #01/01/2026#
```

[4]

```
//Tinokudaishe Mberi
package aicare;
import javax.swing.JOptionPane;
/**
 *
 * @author Tinokduaishe Mberi
 */

//2.1
public class AICare {
    //2.2.1
    public static int totalHours = 0;

    //5.2
    public static int highestHours;
    public static String highestName;


    //2.3
    public static void main(String[] args)
    {

        String input = JOptionPane.showInputDialog("Enter client job name:

        //5.1
        do
        {
            System.out.println(workloadMethod(input));
            System.out.println();
            //5.2
            if(totalHours > highestHours)
            {
                highestHours = totalHours;
                highestName = input;
            }

            input = JOptionPane.showInputDialog("Enter next client job name
        }
        while(!"xxx".equals(input.toLowerCase()));

        //5.3
        JOptionPane.showMessageDialog(null, "Client with the highest total
```

```
");
```

```
: ");
```

```
hours: \nName: "+highestName);
```

```
    }

    public static String workloadMethod(String clientName)
    {
        //3.1
        System.out.println("Simulating status for: "+clientName);

        //3.2
        int dailyHours = 0;

        //3.3
        String day = "";
        totalHours = 0;

        //3.4
        for(int i = 0; i<5;i++)
        {
            //3.4.1
            dailyHours = (int)(Math.random()*5.0)+1;

            //3.4.2
            switch (i)
            {
                case 0 -> day = "Monday";
                case 1 -> day = "Tuesday";
                case 2 -> day = "Wednesday";
                case 3 -> day = "Thursday";
                case 4 -> day = "Friday";
                default -> day = "impossible error, lol";
            }

            //3.4.3
            System.out.println(day+ " - Daily Hours: "+ dailyHours);

            //3.4.4
            totalHours += dailyHours;
        }

        //3.5.1
        if(totalHours > 20)
        {
            totalHours+=3;
            System.out.println("Overtime penalty added (+3). New total: "+
        }
```

```
totalHours);
```

```
        //3.5.2
        calculateBilling();


        //3.5.3
        if(totalHours > 15)
            return "Heavy Load";
        else
            return "Light Load";
    }

    //4
    public static void calculateBilling()
    {
        //4.1
        double weeklyCost = totalHours * 120.00;

        //4.2
        double taxRate = 0, taxAmount = 0;

        //4.3
        if(weeklyCost<2000)
            taxRate = 0.05;
        else
            taxRate = 0.08;

        //4.4
        taxAmount = weeklyCost * taxRate;

        //4.5
        double finalCost = taxAmount + weeklyCost;

        //4.6
        System.out.println("Weekly Cost: "+ weeklyCost);
        System.out.println("Tax Rate: "+ taxRate*100.0 +"%");
        System.out.println("Tax Amount: "+ taxAmount);
        System.out.println("Final Cost: "+ finalCost);

        //4.7
        finalCost = Math.round(finalCost*100.00)/100.00;
        System.out.println("Formatted Final Cost: R" + finalCost);
        System.out.println("-----------------------------------");
    }
```

```
}
```